



# Rogues Gallery VIP - Quantum Computing - Fall 2025

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## Key Research Questions

- How can we leverage simulation data to explore the hardware usage of quantum simulations?
- What are the limits of quantum simulations?
- How may we explore noise mitigation and error correction?
- How accurately do Qiskit simulation noise models reflect actual quantum noise?

## Semester Goals

To continue the exploration of quantum computers and simulations, the team proceeds with work in the following areas:

1. **Data pipeline:** Refactor pipeline for new experiment
2. **Data collection and analysis:** Collect benchmarking, noise simulation, and real data for quantum algorithms.
3. **Quantum simulations:** Create more quantum simulation algorithms (cuQuantum)
4. **Benchmarking:** Determine hardware limitations of quantum simulations

## Onboarding

- Introduce quantum computing fundamentals to new members
- Set up IBM quantum pipeline for all members
- Set up quantum virtual environment for simulation
- Use Slurm to run benchmarking jobs with Intel PCM

## Cleanup Tasks

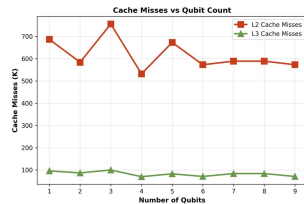
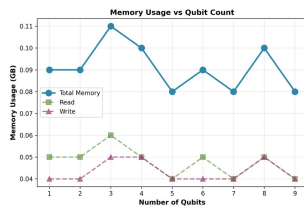
- Refactoring of code to account for new Qiskit versions.
- Improve documentation for running Intel PCM profiling scripts and necessary environment setup
- Refactor simulation files for ease of use

## Areas of Focus

- Noise model vs real data collection
- CPU metric collection, GPU profiling, and more diverse simulation implementation

## Benchmarking/Simulations

- Created cuQuantum implementations of benchmarking algorithms for statevector simulation types
- Collection of CPU profiling data for Simon's algorithm and Bernstein-Vazirani algorithms implemented using cuQuantum statevector simulation type (used frozone1 node)
  - Profiling data indicates that Quantum simulations are memory-bound
- Refactored Qiskit algorithm implementations for ease of use
- Added additional simulation types for Qiskit algorithms: statevector, unitary, density matrix
- Narrowed down issues with profiling simulations alongside using GPU
  - Resolved conda issues but not venv



## Noise Model Experiment

### Updates to Data Pipeline

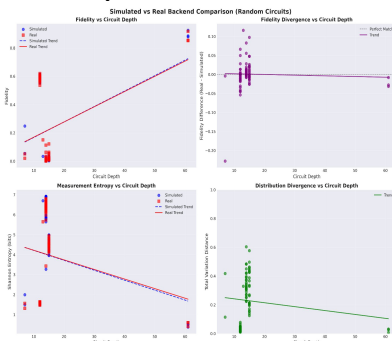
- Running the pipeline now runs both real and simulation and stores results together in a parquet.
- Pipeline updated to account for new Qiskit version

### Experiment:

- Did many runs of simulated with noise models vs real backends
- Automated this flow with crontab

### Conclusion:

The IBM noise models are extremely accurate as circuit fidelity and count distribution is incredibly similar for the simulation and the real backend runs regardless of circuit depth.



## Challenges

- IBM account issues for running pipeline
- Working with old (2+ years) cuQuantum files caused some confusion without clear documentation
- Qiskit constantly changing & deprecating assets so constantly having to refactor
- Nvidia's container for cuquantum does not work well with venv

## Future Work

- **Continuation of Project Work**
  - Continue exploration of IBM error models
  - **New research question:** impact of different types of error on circuit performance?
  - Example error models on Qiskit: coherent unitary error, phase damping error models
  - Implementation of cuQuantum simulation types aside from statevector
  - Improve documentation for future iterations
  - Access to QCUPS?
  - Explore different hardware types
  - Investigate error rates on different hardware
  - Develop more graphical representation of profiling metrics

## References

Intel/Pcm. 31 Oct. 2016. C++. Intel® Corporation, 27 Nov. 2025. GitHub, <https://github.com/intel/pcm>.